

Generalized Predictive Control—Part I. The Basic Algorithm*

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A new member of the family of long-range predictive controllers is shown to be suitable for the adaptive control of processes with varying parameters, dead-time and model-order.

Key Words—Adaptive control; predictive control; LQ control; self-tuning control; nonminimum-phase plant.

Abstract—Current self-tuning algorithms lack robustness to prior choices of either dead-time or model order. A novel method—generalized predictive control or GPC—is developed which is shown by simulation studies to be superior to accepted techniques such as generalized minimum-variance and pole-placement. This receding-horizon method depends on predicting the plant's output over several steps based on assumptions about future control actions. One assumption—that there is a “control horizon” beyond which all control increments become zero—is shown to be beneficial both in terms of robustness and for providing simplified calculations. Choosing particular values of the output and control horizons produces as subsets of the method various useful algorithms such as GMV, EPSAC, Peterka's predictive controller (1984, *Automatica*, 20, 39–50) and Ydstie's extended-horizon design (1984, IFAC 9th World Congress, Budapest, Hungary). Hence GPC can be used either to control a “simple” plant (e.g. open-loop stable) with little prior knowledge or a more complex plant such as nonminimum-phase, open-loop unstable and having variable dead-time. In particular GPC seems to be unaffected (unlike pole-placement strategies) if the plant model is overparameterized. Furthermore, as offsets are eliminated by the consequence of assuming a CARIMA plant model, GPC is a contender for general self-tuning applications. This is verified by a comparative simulation study.

1. INTRODUCTION

ALTHOUGH SELF-TUNING and adaptive control has made much progress over the previous decade, both in terms of theoretical understanding and practical applications, no one method proposed so far is suitable as a “general purpose” algorithm for the stable control of the majority of real processes. To be considered for this role a method must be applicable to:

(1) a nonminimum-phase plant: most continuous-time transfer functions tend to exhibit discrete-time zeros outside the unit circle when sampled at a fast enough rate (see Clarke, 1984);

(2) an open-loop unstable plant or plant with badly-damped poles such as a flexible spacecraft or robots;

(3) a plant with variable or unknown dead-time: some methods (e.g. minimum-variance self-tuners, Åström and Wittenmark, 1973) are highly sensitive to the assumptions made about the dead-time and approaches (e.g. Kurz and Goedecke, 1981) which attempt to estimate the dead-time using operating data tend to be complex and lack robustness;

(4) a plant with unknown order: pole-placement and LQG self-tuners perform badly if the order of the plant is overestimated because of pole/zero cancellations in the identified model, unless special precautions are taken.

The method described in this paper—Generalized Predictive Control or GPC—appears to overcome these problems in one algorithm. It is capable of stable control of processes with variable parameters, with variable dead-time, and with a model order which changes instantaneously provided that the input/output data are sufficiently rich to allow reasonable plant identification. It is effective with a plant which is simultaneously nonminimum-phase and open-loop unstable and whose model is overparameterized by the estimation scheme without special precautions being taken. Hence it is suited to high-performance applications such as the control of flexible systems.

Hitherto the principal applied self-tuning methods have been based on the “Generalized Minimum-Variance” approach (Clarke and Gawthrop, 1975, 1979) and the pole-placement algorithm (Wellstead *et al.*, 1979; Åström and

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Wittenmark, 1980). The implicit GMV self-tuner (so-called because the controller parameters are directly estimated without an intermediate calculation) is robust against model order assumptions but can perform badly if the plant dead-time varies. The explicit pole-placement method, in which a Diophantine equation is numerically solved as a bridging step between plant identification and the control calculation, can cope with variable dead-time but not with a model with overspecified order. Its good behaviour with variable dead-time is due to overparameterization of the numerator dynamics $B(q^{-1})$; this means that the order of the denominator dynamics has to be chosen with great care to avoid singularities in the resolution of the Diophantine identity. The GPC approach being based on an explicit plant formulation can deal with variable dead-time, but as it is a predictive method it can also cope with overparameterization.

The type of plant that a self-tuner is expected to control varies widely. On the one hand, many industrial processes have "simple" models: low order with real poles and probably with dead-time. For some critical loops, however, the model might be more complex, such as open-loop unstable, underdamped poles, multiple integrators. It will be shown that GPC has a readily-understandable *default* operation which can be used for a simple plant without needing the detailed prior design of many adaptive methods. Moreover, at a slight increase of computational time more complex processes can be accommodated by GPC within the basic framework.

All industrial plants are subjected to load-disturbances which tend to be in the form of random-steps at random times in the deterministic case or of Brownian motion in stochastic systems. To achieve offset-free closed-loop behaviour given these disturbances the controller must possess inherent integral action. It is seen that GPC adopts an integrator as a natural consequence of its assumption about the basic plant model, unlike the majority of designs where integrators are added in an *ad hoc* way.

2. THE CARIMA MODEL AND OUTPUT PREDICTION

When considering regulation about a particular operating point, even a non-linear plant generally admits a locally-linearized model:

$$A(q^{-1})y(t) = B(q^{-1})u(t-1) + x(t) \quad (1)$$

where A and B are polynomials in the backward shift operator q^{-1} :

$$\begin{aligned} A(q^{-1}) &= 1 + a_1 q^{-1} + \dots + a_{na} q^{-na} \\ B(q^{-1}) &= b_0 + b_1 q^{-1} + \dots + b_{nb} q^{-nb}. \end{aligned}$$

If the plant has a non-zero dead-time the leading elements of the polynomial $B(q^{-1})$ are zero. In (1), $u(t)$ is the control input, $y(t)$ is the measured variable or output, and $x(t)$ is a disturbance term.

In the literature $x(t)$ has been considered to be of moving average form:

$$x(t) = C(q^{-1})\xi(t) \quad (2)$$

where $C(q^{-1}) = 1 + c_1 q^{-1} + \dots + c_{nc} q^{-nc}$.

In this equation, $\xi(t)$ is an uncorrelated random sequence, and combining with (1) we obtain the CARMA (Controlled Auto-Regressive and Moving-Average) model:

$$\begin{aligned} A(q^{-1})y(t) &= B(q^{-1})u(t-1) \\ &+ C(q^{-1})\xi(t). \end{aligned} \quad (3)$$

Though much self-tuning theory is based on this model it seems to be inappropriate for many industrial applications in which disturbances are non-stationary. In practice, two principal disturbances are encountered: random steps occurring at random times (for example, changes in material quality) and Brownian motion (found in plants relying on energy balance). In both these cases an appropriate model is:

$$x(t) = C(q^{-1})\xi(t)/\Delta \quad (4)$$

where Δ is the differencing operator $1 - q^{-1}$. Coupled with (1) this gives the CARIMA model (integrated moving-average):

$$\begin{aligned} A(q^{-1})y(t) &= B(q^{-1})u(t-1) \\ &+ C(q^{-1})\xi(t)/\Delta. \end{aligned}$$

This model has been used by Tuffs and Clarke (1985) to derive GMV and pole-placement self-tuners with inherent integral action. For simplicity in the development here $C(q^{-1})$ is chosen to be 1 (alternatively C^{-1} is truncated and absorbed into the A and B polynomials; see Part II for the case of general C) to give the model:

$$A(q^{-1})y(t) = B(q^{-1})u(t-1) + \xi(t)/\Delta. \quad (5)$$

To derive a j -step ahead predictor of $y(t+j)$ based on (5) consider the identity:

$$1 = E_j(q^{-1})A\Delta + q^{-j}F_j(q^{-1}) \quad (6)$$

where E_j and F_j are polynomials uniquely defined given $A(q^{-1})$ and the prediction interval j . If (5) is

multiplied by $E_j \Delta q^j$ we have:

$$E_j A \Delta y(t+j) = E_j B \Delta u(t+j-1) + E_j \xi(t+j)$$

and substituting for $E_j A \Delta$ from (6) gives:

$$y(t+j) = E_j B \Delta u(t+j-1) + F_j y(t) + E_j \xi(t+j). \quad (7)$$

As $E_j(q^{-1})$ is of degree $j-1$ the noise components are all in the future so that the optimal predictor, given measured output data up to time t and any given $u(t+i)$ for $i > 1$, is clearly:

$$\hat{y}(t+j|t) = G_j \Delta u(t+j-1) + F_j y(t) \quad (8)$$

where $G_j(q^{-1}) = E_j B$.

Note that $G_j(q^{-1}) = B(q^{-1})[1 - q^{-j}F_j(q^{-1})]/A(q^{-1})\Delta$ so that one way of computing G_j is simply to consider the Z-transform of the plant's step-response and to take the first j terms (Clarke and Zhang, 1985).

In the development of the GMV self-tuning controller only one prediction $\hat{y}(t+k|t)$ is used where k is the assumed value of the plant's dead-time. Here we consider a whole set of predictions for which j runs from a minimum up to a large value: these are termed the minimum and maximum "prediction horizons". For $j < k$ the prediction process $\hat{y}(t+j|t)$ depends entirely on available data, but for $j \geq k$ assumptions need to be made about future control actions. These assumptions are the cornerstone of the GPC approach.

2.1. Recursion of the Diophantine equation

One way to implement long-range prediction is to have a bank of self-tuning predictors for each horizon j ; this is the approach of De Keyser and Van Cauwenberghe (1982, 1983) and of the MUSMAR method (Mosca *et al.*, 1984). Alternatively, (6) can be resolved numerically for E_j and F_j for the whole range of j s being considered. Both these methods are computationally expensive. Instead a simpler and more effective scheme is to use recursion of the Diophantine equation so that the polynomials E_{j+1} and F_{j+1} are obtained given the values of E_j and F_j .

Suppose for clarity of notation $E = E_j$, $R = E_{j+1}$, $F = F_j$, $S = F_{j+1}$ and consider the two Diophantine equations with $\tilde{A}$ defined as $A\Delta$:

$$1 = E\tilde{A} + q^{-j}F \quad (9)$$

$$1 = R\tilde{A} + q^{-(j+1)}S. \quad (10)$$

Subtracting (9) from (10) gives:

$$0 = \tilde{A}(R - E) + q^{-j}(q^{-1}S - F).$$

The polynomial $R - E$ is of degree j and may be split into two parts:

$$R - E = \tilde{R} + r_j q^{-j}$$

so that:

$$\tilde{A}\tilde{R} + q^{-j}(q^{-1}S - F + \tilde{A}r_j) = 0.$$

Clearly then $\tilde{R} = 0$ and also S is given by $Sq(F - \tilde{A}r_j)$.

As $\tilde{A}$ has a unit leading element we have:

$$r_j = f_0 \quad (11a)$$

$$s_i = f_{i+1} - \tilde{a}_{i+1}r_j \quad (11b)$$

for $i = 0$ to the degree of $S(q^{-1})$;

$$\text{and: } R(q^{-1}) = E(q^{-1}) + q^{-j}r_j \quad (12)$$

$$G_{j+1} = B(q^{-1})R(q^{-1}). \quad (13)$$

Hence given the plant polynomials $A(q^{-1})$ and $B(q^{-1})$ and one solution $E_j(q^{-1})$ and $F_j(q^{-1})$ then (11) can be used to obtain $F_{j+1}(q^{-1})$ and (12) to give $E_{j+1}(q^{-1})$ and so on, with little computational effort. To initialize the iterations note that for $j = 1$:

$$1 = E_1\tilde{A} + q^{-1}F_1$$

and as the leading element of $\tilde{A}$ is 1 then:

$$E_1 = 1, \quad F_1 = q(1 - \tilde{A}).$$

The calculations involved, therefore, are straightforward and simpler than those required when using a separate predictor for each output horizon.

3. THE PREDICTIVE CONTROL LAW

Suppose a future set-point or reference sequence $w(t+j)$; $j = 1, 2, \dots$ is available. In most cases $w(t+j)$ will be a constant w equal to the current set-point $w(t)$, though sometimes (as in batch process control or robotics) future variations in $w(t+j)$ would be known. As in the IDCOM algorithm (Richalet *et al.*, 1978) it might be considered that a smoothed approach from the current output $y(t)$ to w is required which is obtainable from the simple

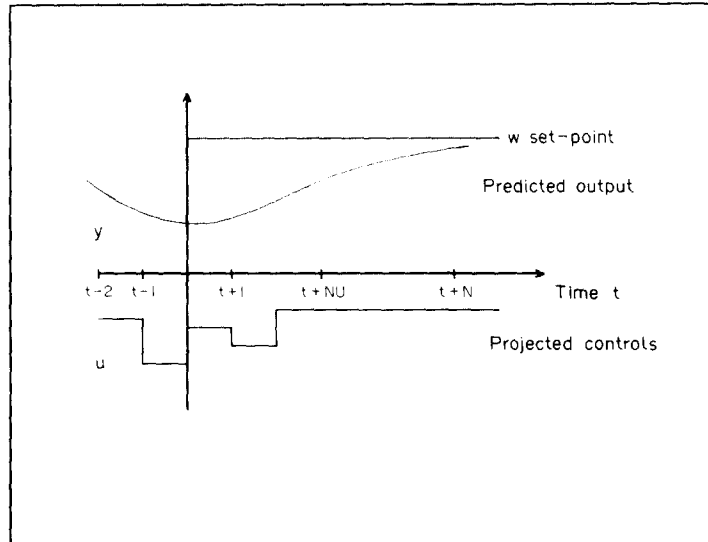


FIG. 1. Set-point, control and outputs in GPC.

first-order lag model:

$$\begin{aligned} w(t) &= y(t) \\ w(t+j) &= \alpha w(t+j-1) \\ &+ (1-\alpha)w \quad j = 1, 2, \dots \end{aligned}$$

where $\alpha \approx 1$ for a slow transition from the current measured-variable to the real set-point w . GPC is capable of considering both constant and varying future set-points.

The objective then of the predictive control law is to drive future plant outputs $y(t+j)$ "close" to $w(t+j)$ in some sense, as shown in Fig. 1, bearing in mind the control activity required to do so. This is done using a receding-horizon approach for which at each sample-instant t :

(1) the future set-point sequence $w(t+j)$ is calculated;

(2) the prediction model of (8) is used to generate a set of predicted outputs $\hat{y}(t+j|t)$ with corresponding predicted system errors $e(t+j) = w(t+j) - \hat{y}(t+j|t)$ noting that $\hat{y}(t+j|t)$ for $j > k$ depends in part on future control signals $u(t+i)$ which are to be determined;

(3) some appropriate quadratic function of the future errors and controls is minimized, assuming that after some "control horizon" further increments in control are zero, to provide a suggested sequence of future controls $u(t+j)$;

(4) the first element $u(t)$ of the sequence is asserted and the appropriate data vectors shifted so that the calculations can be repeated at the next sample instant.

Note that the effective control law is stationary, unlike a fixed-horizon LQ policy. However, in the self-tuned case new estimates of the plant model parameters requires new values for the parameter

polynomials, which means that the fast Diophantine recursion is useful for adaptive control applications of the GPC method.

Consider a cost function of the form:

$$\begin{aligned} J(N_1, N_2) = E \left\{ \sum_{j=N_1}^{N_2} [y(t+j) - w(t+j)]^2 \right. \\ \left. + \sum_{j=1}^{N_2} \lambda(j) [\Delta u(t+j-1)]^2 \right\} \quad (14) \end{aligned}$$

where:

N_1 is the minimum costing horizon;
 N_2 is the maximum costing horizon, and
 $\lambda(j)$ is a control-weighting sequence.

The expectation in (14) is conditioned on data up to time t assuming no future measurements are available (i.e. the set of control signals are applied in open-loop in the sequel). As mentioned earlier, the first control is applied and the minimization is repeated at the next sample. The resulting control law belongs to the class known as Open-Loop-Feedback-Optimal control (Bertsekas, 1976). Appendix A examines the relation between GPC and Closed-Loop-Feedback-Optimal when the disturbance process is autoregressive. It is seen that costing on the control is over all future inputs which affect the outputs included in J . In general N_2 is chosen to encompass all the response which is significantly affected by the current control; it is reasonable that it should at least be greater than the degree of $B(q^{-1})$ as then all states contribute to the cost (Kailath, 1980), but more typically N_2 is set to approximate the rise-time of the plant. N_1 can often be taken as 1; if it is known *a priori* that the dead-time of the plant is at least k sample-intervals then

N_1 can be chosen as k or more to minimize computations. It is found, however, that a large class of plant models can be stabilized by GPC with default values of 1 and 10 for N_1 and N_2 . Part II provides a theoretical justification for these choices of horizon. For simplicity in the derivation, below $\lambda(j)$ is set to the constant λ , N_1 to 1 and N_2 to N : the "output horizon".

Recall that (7) models the future outputs:

$$\begin{aligned} y(t+1) &= G_1 \Delta u(t) + F_1 y(t) + E_1 \xi(t+1) \\ y(t+2) &= G_2 \Delta u(t+1) + F_2 y(t) + E_2 \xi(t+2) \\ &\vdots \\ y(t+N) &= G_N \Delta u(t+N-1) \\ &\quad + F_N y(t) + E_N \xi(t+N). \end{aligned}$$

Consider $y(t+j)$. It consists of three terms: one depending on future control actions yet to be determined, one depending on past known controls together with filtered measured variables and one depending on future noise signals. The assumption that the controls are to be performed in open-loop is tantamount to ignoring the future noise sequence $\{\xi(t+j)\}$ in calculating the predictions. Let $f(t+j)$ be that component of $y(t+j)$ composed of signals which are known at time t , so that for example:

$$\begin{aligned} f(t+1) &= [G_1(q^{-1}) - g_{10}] \Delta u(t) \\ &\quad + F_1 y(t), \quad \text{and} \\ f(t+2) &= q[G_2(q^{-1}) - q^{-1}g_{21} \\ &\quad - g_{20}] \Delta u(t) + F_2 y(t), \\ &\quad \text{etc.} \end{aligned}$$

where $G_i(q^{-1}) = g_{i0} + g_{i1}q^{-1} + \dots$

Then the equations above can be written in the key vector form:

$$\hat{y} = G\tilde{u} + f \quad (15)$$

where the vectors are all $N \times 1$:

$$\begin{aligned} \hat{y} &= [y(t+1), y(t+2), \dots, y(t+N)]^T \\ \tilde{u} &= [\Delta u(t), \Delta u(t+1), \dots, \Delta u(t+N-1)]^T \\ f &= [f(t+1), f(t+2), \dots, f(t+N)]^T. \end{aligned}$$

As indicated earlier, the first j terms in $G_j(q^{-1})$ are the parameters of the step-response and therefore $g_{ij} = g_j$ for $j = 0, 1, 2, \dots < i$ independent of the particular G polynomial.

The matrix G is then lower-triangular of dimension $N \times N$:

$$G = \begin{bmatrix} g_0 & 0 & \dots & 0 \\ g_1 & g_0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ g_{N-1} & g_{N-2} & \dots & g_0 \end{bmatrix}.$$

Note that if the plant dead-time is $k > 1$ the first $k-1$ rows of G will be null, but if instead N_1 is assumed to be equal to k the leading element is non-zero. However, as k will not in general be known in the self-tuning case one key feature of the GPC approach is that a stable solution is possible even if the leading rows of G are zero.

From the definitions of the vectors above and with:

$$w = [w(t+1), w(t+2), \dots, w(t+N)]^T$$

the expectation of the cost-function of (14) can be written:

$$\begin{aligned} J_1 &= E\{J(1, N)\} \\ &= E\{(y - w)^T(y - w) + \lambda \tilde{u}^T \tilde{u}\} \quad (16) \end{aligned}$$

i.e.

$$J_1 = \{(G\tilde{u} + f - w)^T(G\tilde{u} + f - w) + \lambda \tilde{u}^T \tilde{u}\}.$$

The minimization of J_1 assuming no constraints on future controls results in the projected control-increment vector:

$$\tilde{u} = (G^T G + \lambda I)^{-1} G^T (w - f). \quad (17)$$

Note that the first element of $\tilde{u}$ is $\Delta u(t)$ so that the current control $u(t)$ is given by:

$$u(t) = u(t-1) + \tilde{g}^T (w - f) \quad (18)$$

where $\tilde{g}^T$ is the first row of $(G^T G + \lambda I)^{-1} G^T$. Hence the control includes integral action which provides zero offset provided that for a constant set-point $w(t+i) = w$, say, the vector f involves a unit steady-state gain in the feedback path.

Now the Diophantine equation (6) for $q = 1$ gives

$$1 = E_f(1)A(1)\Delta(1) + F_f(1)$$

and as $\Delta(1) = 0$ then $F_f(1) = 1$ so that $f(t+j) = F_j y(t)$ is a signal whose mean value equals that of $y(t)$. Furthermore, defining $F'_j(q^{-1})$ to be $E_f(q^{-1})\tilde{A}(q^{-1})$ gives:

$$\begin{aligned} F_j(q^{-1})y(t) &= (1 - F'_j(q^{-1})\Delta)y(t) \\ &= y(t) - F'_j \Delta y(t) \end{aligned}$$